

SolarExpress Project Outline

Blockchain-based interplanetary ticketing with NASA Eyes-inspired 3D exploration

1. Project Objective

Build a web application where users can explore the solar system in a cinematic 3D interface, choose planets or moons as destinations, and purchase travel tickets using Ethereum.

2. Problem Statement

Conventional booking systems cannot provide transparent on-chain ownership for futuristic space tourism. SolarExpress introduces a wallet-based purchase flow and verifiable ticket proof on blockchain.

3. Proposed Solution

- Interactive 3D solar system map with clickable planets and moons.
- Destination detail panel with travel package and ETH price.
- Wallet connection (MetaMask) and Ethereum payment flow.
- On-chain ticket proof through transaction hash (optional NFT extension).

4. MVP Features

- Fullscreen 3D scene with orbits and object labels.
- Planet selection and camera focus animation.
- Side panel for destination details and booking controls.
- Wallet connect and buy ticket on Sepolia testnet.
- Purchase confirmation with transaction status and hash.

5. Tech Stack

Frontend

- Next.js + React
- Three.js + React Three Fiber
- State management for selection and booking UI

Blockchain

- Solidity smart contract (deployed via Remix)
- wagmi + viem for wallet and contract calls

- Sepolia testnet for demo transactions

6. System Flow

1. User explores the 3D solar system.
2. User selects destination (planet or moon).
3. User chooses ticket package and date.
4. User connects wallet and confirms ETH payment.
5. Contract emits purchase event and returns transaction hash.
6. App displays booking confirmation and ownership proof.

7. Evaluation Mapping

- **Difficulty:** real-time 3D UI plus blockchain transaction integration.
- **Creativity:** space tourism concept with immersive NASA-style navigation.
- **Presentation:** cinematic demo flow from exploration to payment confirmation.
- **Individual:** each member explains one subsystem deeply (3D, contract, UI, integration).

8. Scope and Limitations

- Uses Sepolia testnet, not Ethereum mainnet.
- Destination data and pricing are demo values in MVP.
- Initial release prioritizes interaction and payment proof over complete orbital realism.

9. Future Enhancements

- NFT ticket minting for each booking.
- Dynamic pricing by demand and launch window.
- User booking history and loyalty rewards.
- Expansion to Ethereum Layer 2 for lower gas fees.

10. Foundation Implementation Plan

1. Create Next.js project and install Three.js, React Three Fiber, wagmi, viem.
2. Implement 3D solar system scene with selectable planets.

3. Build destination panel and ticket booking UI.
4. Deploy minimal buyTicket contract from Remix to Sepolia.
5. Integrate wallet connect and contract purchase flow.
6. Add README, screenshots, and demo transaction links for GitHub.

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